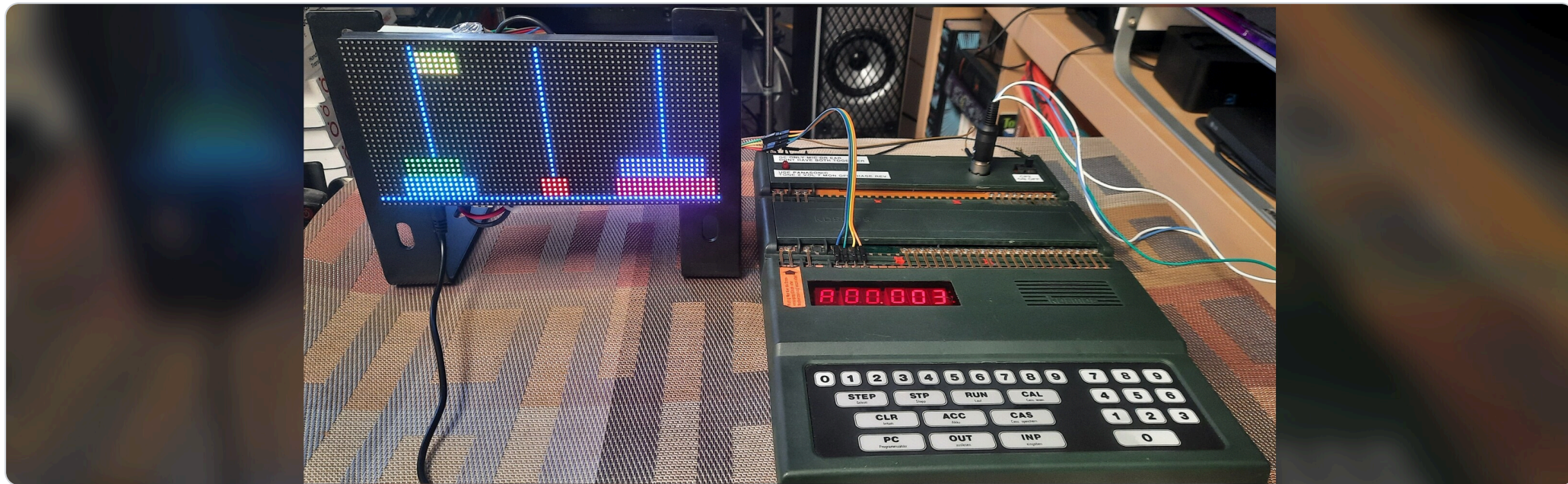


THE TOWERS OF HANOI ON A 1983 MICROCOMPUTER

A **Kosmos CP1** — a 1983 decimal educational computer with only an accumulator and a six-digit display — computes the recursive **Towers of Hanoi** and streams every move to a **64×32 RGB LED matrix** that animates the disks. Michael Wessel's own machine-code original, up to thirteen disks tall.

The legend: 64 golden disks on three needles — when the last is moved, the world ends. At one move a second that is **≈ 585 billion years**, more than 40× the age of the universe.

The feat: an 8-bit **decimal** machine with **no call stack** does true **recursion** — a software stack built by hand in memory — and drives a modern RGB panel over just **four GPIO wires**.



The real rig: a **Kosmos CP1** (right) mid-solve, streaming each move over four jumper wires to an **Arduino Mega 2560** that paints the disks on a **64×32 RGB LED matrix** (left); the program is loaded as a cassette WAV through the CP1's tape input.

01 THE LEGEND OF THE TOWER OF BRAHMA

In a temple said to mark the centre of the world, sixty-four golden disks rest on three diamond needles. Since the day of creation, priests have moved the disks one at a time — never a larger disk upon a smaller — and the day they finish rebuilding the tower on the third needle, the temple crumbles to dust and the world ends.

Rest easy. Rebuilding a 64-disk tower takes $2^{64}-1 = 18,446,744,073,709,551,615$ moves. At one move a second that is **about 585 billion years** — and the universe is only ~13.8 billion years old. The puzzle was invented by the French mathematician **Édouard Lucas** in 1883 and sold as the toy *La Tour d'Hanoi*; the doomsday legend was his marketing.

64

GOLDEN DISKS

 $2^{64}-1$

MOVES REQUIRED

585 bn

YEARS @ 1/SEC

42×

THE UNIVERSE'S AGE

02 THE TRICK — RECURSION ON A MACHINE WITH NO STACK

Towers of Hanoi is the textbook example of **recursion**: to move a stack of N disks, first move the top $N-1$ out of the way, move the big one, then put them back on top.

```
move(N, from→to, via):  
    move(N-1, from→via, to) ; clear the top N-1 onto the spare peg  
    move disk N, from → to ; move the biggest disk  
    move(N-1, via→to, from) ; stack them back on top
```

But the CP1 has only an accumulator and **no call stack** to remember every pending “put them back.” The solver builds one **by hand in memory** — four parallel software stacks (the three pegs plus return addresses) in cells 200–255, grown by a `save_and_push` prologue and unwound by `pop_and_restore`. A 1983 decimal trainer doing honest recursion — up to **thirteen disks**.

03 FROM 1983 SILICON TO A MODERN LED WALL

As it solves, the CP1 streams each move — which disk, from which peg to which — over a **four-bit GPIO protocol** to an **Arduino Mega 2560**, which renders the tower on the **64×32 RGB LED matrix**: disks rainbow-coloured by size, sliding between three pegs on a blue base. It is the very move protocol first built to drive a physical **pan/tilt Hanoi robot** — now painting pixels instead.

And getting the program *onto* the CP1 is painless: a modern **development toolchain** turns readable assembly into a **cassette WAV** you play straight into the tape input — no hand-keying 256 cells on the keypad.



► Watch it · build it

Full solve on the LED matrix, plus the CP1 & Microtronic source, the robot, and the code walk-through.

github.com/lambda-mike/towers-of-hanoi
youtu.be/CkCNBzWRjq4

